

STATISTICS

What is statistics?

- Statistics is the study of data.
- It helps in collecting, organizing, analyzing and interpreting data.
- Earlier, we learned how to show data in tables and graphs.

Measure of Central Tendency:

These give us a central value (a typical value) of the data.

- Mean (Average)
- Median (Middle value)
- Mode (Most frequent value).

* Individual Series (Row data)

⇒ 2, 3, 9, 7, 14, 25, 13

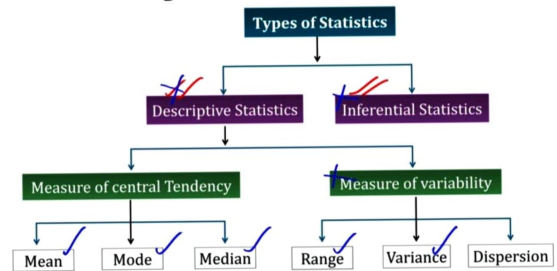
* Discrete data

⇒ $\begin{matrix} x-2 & 4 & 5 & 6 \\ + & - & 7 & 6 & 3 & 1 \end{matrix}$

* Continuous data

⇒ $\begin{matrix} x = 0-10 & 10-20 & 20-30 \\ f = 2 & 4 & 6 \end{matrix}$

Types of Statistics:



Grouped Data v/s Ungrouped Data:

Ungrouped Data: ~~comp~~le

Data that has not been organized into groups. Also called as raw data.

Date	2	3	5	7	8	9
Frequency	8	4	6	7	2	5

Grouped Data: ~~comp~~le

Data that has not been organized into groups (into a frequency distribution).

Date	2-4	5-7	8-10	11-13	14-16	17-18
Frequency	5	6	10	8	4	3

Mean, Median and Mode:

Mean:

The Average. Add all numbers, then divide by how many numbers there are.

Median:

The Middle value in a list when numbers are in order.

Mode:

The number that appears most often in a list.

Types of Series

Raw Data
(Individual Series)

X
7
3
5
4
2
3
2
10

Discrete data

X	f
1	3
2	7
3	8
5	13
6	29
7	1
9	19
10	7

Continuous Data

Class Intervals	f
0-10	23
10-20	17
20-30	19
30-40	157
40-50	239

Mean of Ungrouped data -

Q. Find Mean: 2, 4, 6, 8

Soln: Mean = $\frac{\text{Sum of all observation}}{\text{No. of observation}}$

$$\bar{x} = \frac{2+4+6+8}{4}$$

$$\bar{x} = 20/4$$

$$\bar{x} = 5$$

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$$

'OR'

$$\bar{x} = \frac{\sum fx}{\sum f}$$

Mode of Ungrouped data -

* Most frequently

Ques: Find mode: 10, 5, 3, 10, 7, 2, 3, 10, 1, 19, 10

Ans: Mode = 10

Median of Ungrouped data -

Step (I): Arrange the raw data in ascending or descending order.

Step (II): Observe the number of varieties in the data. Let the number of varieties in the data be n . Then find the median as following:

$n = \text{no. of terms}$

S-I Ascending order

S-II $n = \text{even}$

$n = \text{odd}$

$$\frac{1}{2} \left[\left(\frac{n}{2} \right)^{\text{th}} \text{ term} + \left(\frac{n}{2} + 1 \right)^{\text{th}} \text{ term} \right]$$

Question:

Find the median of the ungrouped data: 15, 18, 10, 6, 14

S-I Ascending order: 6, 10, 14, 15, 18

$n = 5$ Odd

$$\therefore \text{Median} = \left(\frac{n+1}{2} \right)^{\text{th}} \text{ term}$$

$$= \left(\frac{5+1}{2} \right)^{\text{th}} \text{ term}$$

$$= 3^{\text{rd}} \text{ term}$$

Median = 14

KBDT: Median

Odd: 8, 15, 18, 10, 6, 14

A.O $\Rightarrow$ 8, 10, 14, 15, 18

Median = 14

Question:

Find the Median of the raw data:

8, 7, 15, 12, 10, 8, 9, 3

$\Rightarrow$ A.O $\Rightarrow$ 3, 7, 8, 9, 10, 12, 15

$n = 8$ Even

$$\therefore \text{Median} = \frac{1}{2} \left[\left(\frac{n}{2} \right)^{\text{th}} \text{ term} + \left(\frac{n}{2} + 1 \right)^{\text{th}} \text{ term} \right]$$

$$= \frac{1}{2} \left[\left(\frac{8}{2} \right)^{\text{th}} \text{ term} + \left(\frac{8}{2} + 1 \right)^{\text{th}} \text{ term} \right]$$

$$= \frac{1}{2} [4^{\text{th}} \text{ term} + 5^{\text{th}} \text{ term}]$$

$$= \frac{1}{2} [8 + 9] \Rightarrow \frac{17}{2} \Rightarrow 8.5$$

Median

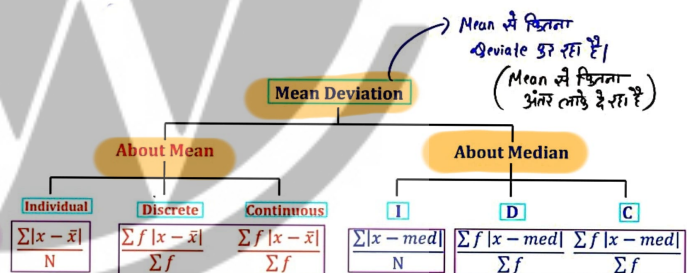
Now by KBDT: Median $\rightarrow$ Even

Q. 8, 7, 15, 12, 10, 8, 9, 3

A.O $\Rightarrow$ 3, 7, 8, 9, 10, 12, 15

$$\Rightarrow \frac{8+9}{2} \Rightarrow \frac{17}{2} \Rightarrow 8.5$$

Median



Q. Find the Mean deviation about the mean for the data

4, 7, 8, 9, 10, 12, 13, 17

Soln: 4, 7, 8, 9, 10, 12, 13, 17

$$\hookrightarrow \text{Mean}(\bar{x}) = \frac{4+7+8+9+10+12+13+17}{8}$$

$$\bar{x} = 80/8 = 10$$

Deviation $\rightarrow$ अंतर/कमी $\frac{\sum |x - \bar{x}|}{N}$

$\Rightarrow |4-10|, |7-10|, |18-10|, |19-10|, |10-10|, |12-10|, |13-10|, |17-10|$
 $\Rightarrow \{6, 3, 2, 1, 0, 2, 3, 7\}$ wala data $\rightarrow$ Deviation

$$M_{\text{mean}} = \frac{6+3+2+1+0+2+3+7}{8}$$

$$= \frac{24}{8}$$

$$= 3 \text{ (final Answer)}$$

$$M_{\text{mean}} = \frac{\sum |x - \bar{x}|}{N}$$

Q. Find the Mean deviation about the Mean for the data.

x_i	5	10	15	20	25
x_f	7	4	6	3	5

x_i	f_i	$f_i x_i$	$ x - \bar{x} $	$\sum f_i x - \bar{x} $
5	7	35	9	63
10	4	40	4	16
15	6	90	1	6
20	3	60	6	18
25	5	125	11	55
$\sum f_i = 25$		$\sum f_i x_i = 350$		$\sum f_i x - \bar{x} = 158$

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}$$

$$= \frac{350}{25} = 14$$

Deviation Mean:

$$\frac{\sum f_i |x - \bar{x}|}{\sum f_i}$$

$$\Rightarrow \frac{158}{25} = 6.32$$

Ans.

Q. Find the Mean deviation about the median for the data.
 13, 17, 16, 14, 11, 13, 10, 16, 11, 18, 12, 17.

Sol: Median

$$A.O \Rightarrow 10, 11, 11, 12, 13, 13, 14, 16, 16, 17, 17, 18$$

$$13, 13, 14, 16, 16, 17, 17, 18$$

$$\hookrightarrow \text{Median} = \frac{13+14}{2}$$

$$= \frac{27}{2}$$

$$\text{Median} = 13.5$$

$$\frac{\sum |x - \text{med}|}{N}$$

$$\text{Mean Median} = \frac{++}{N}$$

$$= \frac{28}{12}$$

$$M_{\text{median}} = 2.33$$

x	$ x - \bar{x} $
13	0.5
14	3.5
16	2.5
14	0.5
11	2.5
13	0.5
10	3.5
16	2.5
11	2.5
18	4.5
12	1.5
17	3.5
$\sum x - \bar{x} = 28$	

Q. Find the mean deviation about the median for the data.

x_i	5	7	9	10	12	15
x_f	8	6	2	2	2	6

x_i	f_i	C.F	$ x - \text{med} $	$f_i x - \text{med} $
5	8	8	2	16
7	6	14	0	0
9	2	16	2	4
10	2	18	3	6
12	2	20	5	10
15	6	26	8	48

$$\sum f_i$$

$$\sum f_i |x - \text{med}| = 84$$

Shortcut: (KBDT)

$$\hookrightarrow \text{Median} = \frac{\text{Total } f}{2} \left(\frac{N}{2} \right)$$

$$= \frac{26}{2} \Rightarrow (13) \text{ Median class.}$$

$$M.D(\text{Median}) = \frac{\sum f_i |x - \text{med}|}{\sum f_i}$$

$$= \frac{84}{26} = 3.23$$

$$= 3.23 \text{ Ans.}$$

Variance & Standard Deviation.

⊕ Variance:

It shows how far the numbers are from the average (mean).

- If variance is small → numbers are close to the mean
- If variance is large → numbers are spread out

⊕ Standard Deviation:

It is the square root of variance. It tells how much the data values differ from the average - but in the same units as the data.

⊕ In very simple words:

Variance: Average of how far each number is from the mean
Standard Deviation: shows how spread out the data is (in same units)

$$\text{Var} = \sigma^2$$

$$\text{SD} = \sigma$$

$$\text{SD} = \sqrt{\text{Var}}$$

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Example:

$$\text{Variance} = 24.2$$

$$\text{Standard deviation} = \sqrt{24.2}$$

$$(\sigma)^2 = 24.2$$

$$(\sigma) = \sqrt{24.2}$$

SD

Q. Find the mean and variance for each of the data:

$$6, 7, 10, 12, 13, 4, 8, 12$$

$$\text{Mean} = \frac{6+7+10+12+13+4+8+12}{8}$$

$$= \frac{72}{8} = 9$$

$$* \text{Mean}(\bar{x}) = \frac{\sum x}{N}$$

$(x - \bar{x})$ After denat

$$* \text{Variance} = \frac{\sum x^2}{N}$$

$$= \frac{74}{8}$$

$$(\sigma)^2 = 9.25 \text{ Ans}$$

x	(x - $\bar{x}$)	x^2
6	(6-9) = -3	(-3) ² = 9
7	(7-9) = -2	(-2) ² = 4
10	(10-9) = 1	(1) ² = 1
12	(12-9) = 3	(3) ² = 9
13	(13-9) = 4	(4) ² = 16
4	(4-9) = -5	(-5) ² = 25
8	(8-9) = -1	(-1) ² = 1
12	(12-9) = 3	(3) ² = 9
$\sum x = 72$		$\sum x^2 = 74$

Q. Find the Mean and standard deviation:

x_i	60	61	62	63	64	65	66	67	68
x_f	2	1	12	29	25	12	10	4	5

x_i	f_i	$f_i x_i$	$(x - \bar{x})$	$x^2 = (x - \bar{x})^2$	$f_i x^2$
60	2	120	-4	16	32
61	1	61	-3	9	9
62	12	744	-2	4	48
63	29	1827	-1	1	29
64	25	1600	0	0	0
65	12	780	1	1	12
66	10	660	2	4	40
67	4	268	3	9	36
68	5	340	4	16	80

$$\sum f = 100$$

$$\sum f_i x_i = 6400$$

$$286$$

$$\text{Mean} = \frac{\sum f_i x_i}{\sum f_i}$$

$$= \frac{6400}{100}$$

$$= 64$$

$$\sigma^2 = \frac{\sum f x^2}{\sum f}$$

$$= \frac{286}{100}$$

$$\sigma^2 = 2.86$$

$$\text{Standard deviation} = \sqrt{2.86} = 1.6$$

Q. Calculate the Mean, Variance, and standard deviation for the following distribution.

Class	30-40	40-50	50-60	60-70	70-80	80-90	90-100
Frequency	3	7	12	15	8	3	2

Class	Frequency (f)	Mid-point (x)	$y_i = \frac{x_i - 65}{10}$	y_i^2	$f_i y_i$	$f_i y_i^2$
30-40	3	35	-3	9	-9	27
40-50	7	45	-2	4	-14	28
50-60	12	55	-1	1	-12	12
60-70	15	65	0	0	0	0
70-80	8	75	1	1	8	8
80-90	3	85	2	4	6	12
90-100	2	95	3	9	6	18
N = 50					-15	105

$$(35-62)^2 = 729$$

$$\begin{aligned} \text{Mean}(\bar{x}) &= \frac{\sum f_i x_i}{\sum f_i} \\ &= \frac{3100}{50} \\ \bar{x} &= 62 \end{aligned}$$

$$\sigma^2 = 201$$

$$S.D = 14.18$$

$$\text{Variance}(\sigma)^2 = \frac{1}{N} \sum_{i=1}^7 f_i (x_i - \bar{x})^2$$

$$= \frac{1}{50} \times 10050$$

$$\sigma^2 = 201$$

$$S.D = \sqrt{201}$$

$$S.D = 14.18$$

$$\begin{array}{r} 25 \quad 24 \\ 9 \quad 24 \\ \hline 125 \quad 96 \\ 14. \\ \hline 1 \quad 0201 \\ 1 \quad 1 \\ \hline 24 \quad 101 \\ 4 \quad -96 \\ \hline 28 \quad 500. \end{array}$$

8. Calculate mean, variance and standard deviation for the following distribution. [Shortcut Method]

Class	30-40	40-50	50-60	60-70	70-80	80-90	90-100
freq	3	7	12	15	8	3	2.

Class	Frequency (f _i)	Mid-point (x _i)	f _i x _i	(x _i - $\bar{x}$) ²	f _i (x _i - $\bar{x}$) ²
30-40	3	35	105	729	2187
40-50	7	45	315	289	2023
50-60	12	55	660	49	588
60-70	15	65	975	9	135
70-80	8	75	600	169	1352
80-90	3	85	255	529	1587
90-100	2	95	190	1089	2178
$\sum f_i$	50		3100		10050

Mean($\bar{x}$)
assumed
Mean
($\bar{x}$) = 65

$$\begin{aligned} \sigma^2 &= \frac{h^2}{N^2} [N \sum f_i d_i^2 - (\sum f_i d_i)^2] \\ &= \frac{(10)^2}{(50)^2} [50(105) - (-15)^2] \end{aligned}$$

$$= \frac{10 \times 10}{50 \times 50} [5250 - 225]$$

$$= \frac{10 \times 10}{50 \times 50} \times 5025$$